



Chemical Theory II

Statistical

Thermodynamics

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The course is developed based on the
Cambridge chemistry courses MELT

Summary of last lecture

- Symmetry of wavefunctions
 - Nuclei have spin.
Interchange of identical bosons leaves Ψ unchanged.
Interchange of identical fermions changes the sign of Ψ .
- Overall wavefunction must have the right symmetry.
- Swapping nuclear position and spin coordinates:
 - C_2 : rotate entire molecule (odd J states change sign)
 - i_{elec} : invert electrons (ungerade terms change sign)
 - σ_{elec} : reflect electrons (Σ^- terms change sign)
 - p_{nuc} : swap nuclear spins (anti-symmetric spin functions change sign)

Summary of last lecture

- Symmetry of wavefunctions
 - Some combinations of J and nuclear spin functions are not allowed.
- Consequences
 - Different statistical weights of even/odd rotational J states
 - Alternating intensity or missing lines in Raman spectra
 - Ortho and para hydrogen
- Nuclear spin statistics
 - Symmetry factor σ in $q_{\text{rot}} = kT / \sigma B$
 - Half the J states are missing in homonuclear diatomics.
 - In general, σ = number of superimposable rotated structures.
 - Some combinations of J and nuclear spin function are not allowed.

Evaluation of the rotational partition function

- In earlier lectures, we pointed out that we could find the molecular partition function by multiplying together the partition functions for different sets of energy levels. For example:

$$q = q_{\text{trans}} q_{\text{NS}} q_{\text{rot}} q_{\text{vib}} q_{\text{elec}}$$

where q_{NS} is the nuclear spin partition function.

- *This separation only works if there are no restrictions on the energy levels which molecules can occupy* i.e. any nuclear spin state can go with any rotational state. Under these circumstances the sums needed to find each of the separate contributions to q can be computed independently of one another.

Evaluation of the rotational partition function

- It is the nuclear spin and rotational contributions which are entangled, so let us for the moment just consider the product $q_{\text{NS}} q_{\text{rot}}$. In a heteronuclear diatomic there are no complications, and we can evaluate these two partition functions separately. As the levels are degenerate we must compute q :

$$q = \sum_j g_j \exp\left(\frac{-\varepsilon_j}{kT}\right)$$

- If the two nuclei have spins I_1 and I_2 , then, as explained previously, there are $(2I_1 + 1)(2I_2 + 1)$ nuclear spin states, *all of which are degenerate and which we define to have energy zero.*

Evaluation of the rotational partition function

- It is easy to convince ourselves that in the limit $kT \gg B$, where many terms contribute to the sums, the sum over just the even or odd J -states is equal to half the sum over all the J -states.

$$\begin{aligned} \sum_{\text{even } J} (2J+1) \exp\left(\frac{-BJ(J+1)}{kT}\right) &= \sum_{\text{odd } J} (2J+1) \exp\left(\frac{-BJ(J+1)}{kT}\right) \\ \sum_{\text{even } J \text{ or odd } J} (2J+1) \exp\left(\frac{-BJ(J+1)}{kT}\right) &= \frac{1}{2} \sum_{\text{all } J} (2J+1) \exp\left(\frac{-BJ(J+1)}{kT}\right) \\ &= \frac{kT}{2B} \end{aligned}$$

where we have used the result for the rotational partition function from $q_{\text{rot}} = \frac{kT}{\sigma B}$.

- So, *in this limit, for a homonuclear diatomic we have*

$$\begin{aligned} \text{homonuclear: } q_{\text{NS}} q_{\text{rot}} &= g_{\text{NS,even}} \frac{kT}{2B} + g_{\text{NS,odd}} \frac{kT}{2B} \\ &= (g_{\text{NS,even}} + g_{\text{NS,odd}}) \frac{kT}{2B} \end{aligned}$$

whereas for a heteronuclear diatomic we have

$$\text{heteronuclear: } q_{\text{NS}} q_{\text{rot}} = g_{\text{NS}} \frac{kT}{B}$$

- The total number of nuclear spin states is the same in each case (assuming that the nuclear spins are the same), so $g_{\text{NS}} = (g_{\text{NS,even}} + g_{\text{NS,odd}})$. It therefore follows that for a homonuclear diatomic the product $q_{\text{NS}} q_{\text{rot}}$ is one half of that for an equivalent heteronuclear diatomic.

Evaluation of the rotational partition function

- So, leaving out the nuclear spin part we have:

$$\text{heteronuclear: } q_{\text{rot}} = \frac{kT}{B} \quad \text{homonuclear: } q_{\text{rot}} = \frac{kT}{2B}$$

- *This, then, is the origin of the symmetry parameter, σ , introduced in earlier lectures. In summary, in a homonuclear diatomic there are restrictions on which rotational states can combine with which nuclear spin states. As a result, fewer combinations of levels are available to the molecule and so the partition function is less than in an equivalent heteronuclear diatomic where no such restrictions exist.*
- Similar arguments can be put forward for more complex molecules. For symmetrical linear molecules (such as CO_2 and HCCH) $\sigma = 2$ by analogy with the case of a homonuclear diatomic. For non-linear molecules it turns out that σ is equal to the number of equivalent positions into which the molecule can be rotated, so it is 2 for H_2O and 3 for NH_3 .

Course outline

- Chemical equilibrium

Course outline

- One of the most remarkable applications of statistical thermodynamics is the calculation of equilibrium constants. We will show in this section how it is possible to calculate these just from a knowledge of the energy levels of the reacting molecules. *The results in this section apply only to indistinguishable molecules.*

Chemical potential

- As we saw in this lecture, we need to introduce the concept of *chemical potential*, μ , to deal with the thermodynamics of mixtures. The key point is that the Gibbs energy of a mixture can be computed from

$$G = \sum_i n_i \mu_i$$

where n_i is the number of moles of substance i whose chemical potential is μ_i .

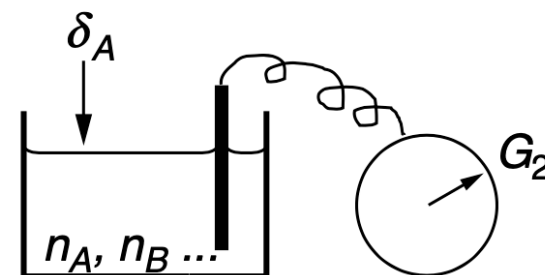
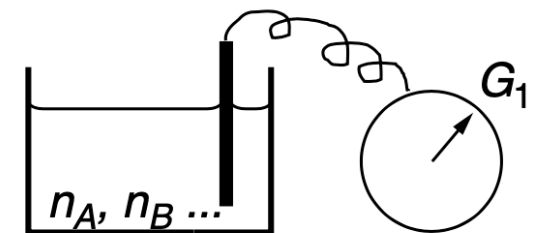
- This relationship shows us that *the Gibbs energy of a mixture depends on its composition* i.e. how many moles of each substance there are present. *In addition, the chemical potentials themselves also depend on the composition of the mixture.*

Chemical potential

- The chemical potential was defined before for substance i as

$$\mu_i = \left(\frac{\partial G}{\partial n_i} \right)_{p, T, n_j}$$

- In words, *the chemical potential of i is the change in G when an infinitesimal number of moles of i , dn , is added to the mixture such that the composition remains fixed, all at fixed temperature and pressure.*



$$\mu_A = \frac{G_2 - G_1}{\delta_A}$$

Visualization of the meaning of chemical potential. A small amount of substance A is added to a mixture and the increase in G (indicated here on a 'G meter' is recorded. The composition of the mixture remains constant as only an infinitesimal quantity is added.

Chemical potential

- With the definition of chemical potential, we can work out how **G** changes when the amounts of each component are varied:

$$dG = \sum_i \mu_i dn_i \quad \text{const. } T \text{ and } p$$

- In this lecture we used this to find *the relationship between the equilibrium constant, **K**, and the standard Gibbs energy change of a reaction, $\Delta_r G^\circ$:* **$\Delta_r G^\circ = -RT \ln K$** . It turns out there are a number of equivalent definitions of the chemical potential, all of which result in the same properties.

Chemical potential

- The one which we will find most useful is

$$\mu_i = \left(\frac{\partial A}{\partial N_i} \right)_{V,T,N_j}$$

where N_i is the number of molecules (or atoms) rather than the number of moles; note the conditions of constant temperature and volume.

- For a single substance, A is given by (for indistinguishable particles)

$$A = -NkT \ln q + kT (N \ln N - N)$$

- Hence, the chemical potential is

$$\begin{aligned} \mu &= \left(\frac{\partial A}{\partial N} \right)_{T,V} = \frac{\partial}{\partial N} (-NkT \ln q + kT (N \ln N - N)) \\ &= -kT \ln q + kT \ln N + \frac{NkT}{N} - kT \\ &= -kT \ln \frac{q}{N}. \end{aligned}$$

Chemical potential

- In the case that we have a mixture of substances which are non-interacting, we expect a similar expression for the chemical potential of each substance. The argument is that, as there are no interactions, each substance contributes to the total **A** in a way which is independent of the others. So, we write

$$\mu_i = -kT \ln \frac{q_i}{N_i}$$

where **q_i** is the partition function of molecule (or atom) *i* and **N_i** is the number of such molecules present.

Choice of energy zero

$$q = q_{\text{trans}} q_{\text{NS}} q_{\text{rot}} q_{\text{vib}} q_{\text{elec}}$$

- In earlier slides we discussed how the choice of energy zero affected the value of q and hence the thermodynamic functions, such as A . If we are going to consider chemical reactions we need to use the same energy zero for all the molecules, otherwise our calculations will be in error.
- *For translational and rotational energy levels the ground states have zero energy for all molecules;* so, without further effort, these sets of energy levels are referred to the same level.

Choice of energy zero

$$q = q_{\text{trans}} q_{\text{NS}} q_{\text{rot}} q_{\text{vib}} q_{\text{elec}}$$

- *For vibrational levels we can measure the energies relative to the bottom of the potential energy well*, leading to the form of q_{vib} :

$$q_{\text{vib}} = \frac{\exp\left(\frac{-\theta_{\text{vib}}}{2T}\right)}{1 - \exp\left(\frac{-\theta_{\text{vib}}}{T}\right)}$$

- The potential energy well is in fact a plot of the electronic energy as a function of the displacement of the bond from its equilibrium length. *At the minimum the energy is not zero but corresponds to the electronic energy of the molecule at its equilibrium geometry. So, by using the formula above we are measuring the vibrational energy levels relative to the (ground state) electronic energy of the molecule.*

Choice of energy zero

$$q = q_{\text{trans}} q_{\text{NS}} q_{\text{rot}} q_{\text{vib}} q_{\text{elec}}$$

- As was noted before, different atoms and molecules have different energies of their electronic ground states; in order to compute q_{elec} we need to know these energies. However, *in the calculation of equilibrium constants it turns out that all we need to know are the changes in these energies between reactants and products – their absolute values are not important.*
- We are therefore at liberty to express these energies relative to some arbitrary origin, and *here the origin we will choose will be the energy of the dissociated atoms.*

Choice of energy zero

- Following the earlier lecture we will write the Helmholtz energy as

$$A = -NkT \ln q + kT (N \ln N - N) + N\varepsilon^0$$

and so the chemical potential is

$$\begin{aligned}\mu &= \left(\frac{\partial A}{\partial N} \right)_{T,V} \\ &= -kT \ln \frac{q}{N} + \varepsilon^0\end{aligned}$$

- For species i in an ideal mixture this becomes

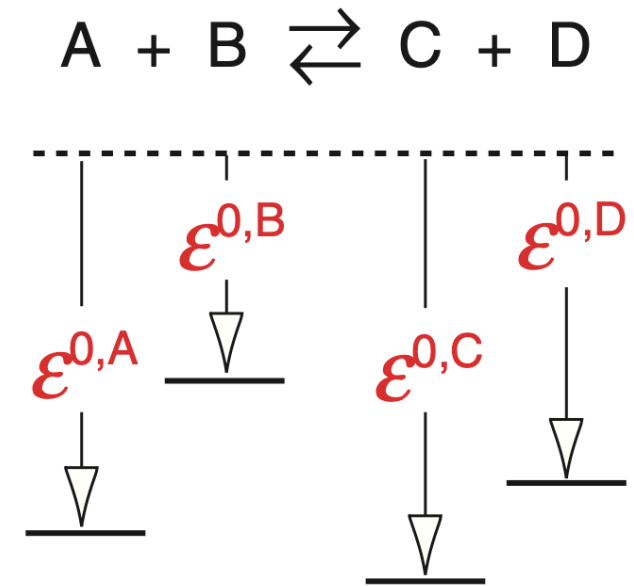
$$\mu_i = -kT \ln \frac{q_i}{N_i} + \varepsilon^{0,i}.$$

Choice of energy zero

$$\mu_i = -kT \ln \frac{q_i}{N_i} + \epsilon^{0,i}.$$

- where $\epsilon^{0,i}$ is the energy of the electronic ground state of species i measured relative to the energy of the dissociated atoms and where it is understood that:

(1) *the vibrational partition function is computed using energies measured from the bottom of the potential energy well* i.e. using the form of q_{vib} ; and (2) *to compute the electronic partition function, the energy of the electronic ground state is taken as zero.*



Visualization of the use of a common energy zero (shown by the dashed line) for a chemical reaction. The electronic ground states of the reactants and products are shown by the short horizontal lines. In this case all of the ground states are below the energy zero, so the ϵ^0 terms are all negative.

Equilibrium constants: Calculation of K_c

- Consider a general chemical equilibrium



where ν_A etc. are the stoichiometric coefficients. In previous slides it was shown that at equilibrium $\Delta_r G = 0$, where $\Delta_r G$ is given by

$$\Delta_r G = \nu_M \mu_M + \nu_N \mu_N - \nu_A \mu_A - \nu_B \mu_B$$

- We will now write each of the chemical potentials using the expression for μ_i in terms of q_i . However, before doing this we will make a slight modification to the expression for μ_i :

$$\mu_i = -kT \ln \frac{q_i}{N_i} + \varepsilon^{0,i}.$$

Equilibrium constants: Calculation of K_c

- Previously, we saw *that of all the contributions to the partition function only one, the translational partition function, contains a volume term*. For present purposes it will suit us to factor this volume out and write $q = f V$
- where f is the volume independent partition function. In practice we compute f in exactly the same way as we compute q except that the volume term is left out of q_{trans} :

$$\begin{aligned} f &= \frac{q}{V} = \frac{q_{\text{trans}}}{V} q_{\text{rot}} q_{\text{vib}} q_{\text{elec}} \\ &= \left(\frac{2\pi mkT}{h^2} \right)^{\frac{3}{2}} q_{\text{rot}} q_{\text{vib}} q_{\text{elec}} \end{aligned}$$

- In terms of the volume independent partition function, the previous expression of the chemical potential can be written

$$\begin{aligned} \mu_i &= -kT \ln \frac{f_i V}{N_i} + \varepsilon^{0,i} \\ &= -kT \ln \frac{f_i}{c_i} + \varepsilon^{0,i} \end{aligned}$$

where $c_i = N_i/V$, the concentration of species i expressed in molecules (or atoms) per unit volume (m^3 in SI).

Equilibrium constants: Calculation of K_c

$$\begin{aligned}\mu_i &= -kT \ln \frac{f_i V}{N_i} + \varepsilon^{0,i} \\ &= -kT \ln \frac{f_i}{c_i} + \varepsilon^{0,i}\end{aligned}$$

$$\Delta_r G = \nu_M \mu_M + \nu_N \mu_N - \nu_A \mu_A + \nu_B \mu_B$$

- This expression for μ_i is then substituted into the equation of $\Delta_r G$ ($\Delta_r G = 0$) to give

$$\begin{aligned}&\nu_M \left(-kT \ln \frac{f_M}{c_M} + \varepsilon^{0,M} \right) + \nu_N \left(-kT \ln \frac{f_N}{c_N} + \varepsilon^{0,N} \right) \\ &- \nu_A \left(-kT \ln \frac{f_A}{c_A} + \varepsilon^{0,A} \right) - \nu_B \left(-kT \ln \frac{f_B}{c_B} + \varepsilon^{0,B} \right) = 0\end{aligned}$$

- Rearrangement of this expression gives

$$kT \ln \frac{c_M^{\nu_M} c_N^{\nu_N}}{c_A^{\nu_A} c_B^{\nu_B}} = kT \ln \frac{f_M^{\nu_M} f_N^{\nu_N}}{f_A^{\nu_A} f_B^{\nu_B}} - \nu_M \varepsilon^{0,M} - \nu_N \varepsilon^{0,N} + \nu_A \varepsilon^{0,A} + \nu_B \varepsilon^{0,B}$$

Equilibrium constants: Calculation of K_c

$$kT \ln \frac{c_M^{\nu_M} c_N^{\nu_N}}{c_A^{\nu_A} c_B^{\nu_B}} = kT \ln \frac{f_M^{\nu_M} f_N^{\nu_N}}{f_A^{\nu_A} f_B^{\nu_B}} - \nu_M \epsilon^{0,M} - \nu_N \epsilon^{0,N} + \nu_A \epsilon^{0,A} + \nu_B \epsilon^{0,B}$$

- Division by kT and changing to exponentials gives

$$\frac{c_M^{\nu_M} c_N^{\nu_N}}{c_A^{\nu_A} c_B^{\nu_B}} = \frac{f_M^{\nu_M} f_N^{\nu_N}}{f_A^{\nu_A} f_B^{\nu_B}} \exp(-\Delta \epsilon_0 / kT)$$

where $\Delta \epsilon_0$ is the difference in the ground state electronic energy levels between products and reactants measured relative to a common energy zero:

$$\Delta \epsilon_0 = \underbrace{\nu_M \epsilon^{0,M} + \nu_N \epsilon^{0,N}}_{\text{products}} \underbrace{- \nu_A \epsilon^{0,A} - \nu_B \epsilon^{0,B}}_{\text{minus reactants}}$$

Equilibrium constants: Calculation of K_c

$$\frac{c_M^{\nu_M} c_N^{\nu_N}}{c_A^{\nu_A} c_B^{\nu_B}} = \frac{f_M^{\nu_M} f_N^{\nu_N}}{f_A^{\nu_A} f_B^{\nu_B}} \exp(-\Delta\epsilon_0/kT)$$

- *The quantity on the left of this equation is almost, but not quite, an equilibrium constant.* Recall that we have to define the equilibrium constant in such a way that it is dimensionless. Before, we did this by dividing each partial pressure by the standard pressure. Here, we divide the concentrations by the standard concentration, c^0 (which in molecular units will be one molecule per m^3). *So the equilibrium constant in terms of concentrations, K_c , is*

$$K_c = \frac{\left(\frac{c_M}{c^0}\right)^{\nu_M} \left(\frac{c_N}{c^0}\right)^{\nu_N}}{\left(\frac{c_A}{c^0}\right)^{\nu_A} \left(\frac{c_B}{c^0}\right)^{\nu_B}} = \frac{c_M^{\nu_M} c_N^{\nu_N}}{c_A^{\nu_A} c_B^{\nu_B}} \left(\frac{1}{c^0}\right)^{\nu_M + \nu_N - \nu_A - \nu_B} = \frac{c_M^{\nu_M} c_N^{\nu_N}}{c_A^{\nu_A} c_B^{\nu_B}} \left(\frac{1}{c^0}\right)^{\Delta\nu}$$

where $\Delta\nu = \nu_M + \nu_N - \nu_A - \nu_B$, the change in number of moles in going from reactants to products. Then we know by combining these two equations:

$$K_c (c^0)^{\Delta\nu} = \frac{f_M^{\nu_M} f_N^{\nu_N}}{f_A^{\nu_A} f_B^{\nu_B}} \exp(-\Delta\epsilon_0/kT)$$

Equilibrium constants: Calculation of K_c

- So the final expression for K_c is

$$K_c = \frac{f_M^{\nu_M} f_N^{\nu_N}}{f_A^{\nu_A} f_B^{\nu_B}} \left(\frac{1}{c^\circ} \right)^{\Delta \nu} \exp(-\Delta \epsilon_0 / kT)$$

- If we check this we will find that the LHS and RHS are both dimensionless, as required (recall that the dimensions of f are m^{-3}). *Essentially K_c in this equation is measured relative to a standard state of 1 molecule m^{-3} , compare this to K_p where the standard state is 1 bar. This equation is a practical relationship for calculating equilibrium constants from partition functions.*
- $\Delta \epsilon_0$ can also be described as the energy change of the reaction at absolute zero. *It must be expressed in terms of energy per molecule (not per mole).*

Equilibrium constants: Calculation of K_p

- Previously it was shown that

$$\Delta_r G^\circ = -RT \ln K_p$$

where K_p is the equilibrium constant expressed in terms of partial pressures and $\Delta_r G^\circ$ is the standard Gibbs energy change for the reaction given in terms of standard chemical potentials, μ_i° :

$$\Delta_r G^\circ = \nu_M \mu_M^0 + \nu_N \mu_N^0 - \nu_A \mu_A^0 - \nu_B \mu_B^0$$

- This time, the standard chemical potential is for one mole (not one molecule) of substance at the standard pressure (1 bar, 10^5 N m^{-2})*

$$\begin{aligned} \mu_i^0 &= N_A \left(-kT \ln \frac{q_i^\circ}{N_A} + \epsilon^{0,i} \right) \\ &= -RT \ln \frac{q_i^\circ}{N_A} + N_A \epsilon^{0,i} \end{aligned}$$

The units of N_A are mol^{-1}

Equilibrium constants: Calculation of K_p

- We have had to multiply by Avogadro's constant as our expression for chemical potential is for one molecule; q° is the molecular partition function in the standard state. *The only part of the partition function that is affected by the specification of the standard state is the volume in the translational partition function.* For the standard state, this volume must be the molar volume at standard pressure, V_m° :

$$\begin{aligned} q^\circ &= q_{\text{trans}} q_{\text{rot}} q_{\text{vib}} q_{\text{elec}} \\ &= \left(\frac{2\pi mkT}{h^2} \right)^{\frac{3}{2}} V_m^\circ q_{\text{rot}} q_{\text{vib}} q_{\text{elec}} \\ &= \left(\frac{2\pi mkT}{h^2} \right)^{\frac{3}{2}} \left(\frac{RT}{p^\circ} \right) q_{\text{rot}} q_{\text{vib}} q_{\text{elec}} \end{aligned}$$

where p° is the standard pressure.

- From this definition it is easy to see that *the units of q° are mol^{-1} , so that q° / N_A is dimensionless*, as required in the equation of μ_i° . *In evaluating q° it is important to keep close track of the units: in SI m must be in kg (not amu) and p° in N m^{-2} .*

Equilibrium constants: Calculation of K_p

$$\begin{aligned}\mu_i^0 &= N_A \left(-kT \ln \frac{q_i^\circ}{N_A} + \varepsilon^{0,i} \right) \\ &= -RT \ln \frac{q_i^\circ}{N_A} + N_A \varepsilon^{0,i}\end{aligned}\quad \Delta_r G^\circ = -RT \ln K_p$$

- Using the left equation in the expression for $\Delta_r G^\circ$ and then substituting this in the right equation gives, after some algebra,

$$K_p = \frac{(q_M^\circ)^{\nu_M} (q_N^\circ)^{\nu_N}}{(q_A^\circ)^{\nu_A} (q_B^\circ)^{\nu_B}} (N_A)^{-\Delta \nu} \exp \left(-\frac{\Delta \varepsilon_0}{kT} \right)$$

where, as before, $\Delta \nu = \nu_M + \nu_N - \nu_A - \nu_B$.

Interpretation

- To get a feeling for what relationships like K_c and K_p mean it is useful to think about a very simple equilibrium



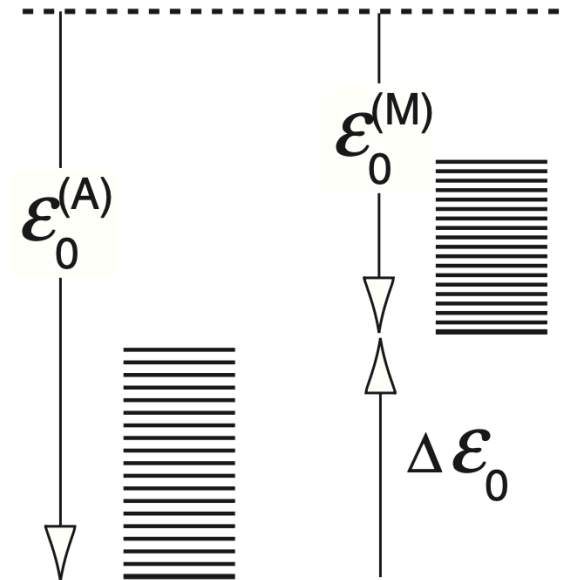
for which K_c is

$$K_c = \frac{f_M}{f_A} \exp(-\Delta\epsilon_0/kT)$$

where

$$\Delta\epsilon_0 = \epsilon^{0,M} - \epsilon^{0,A}$$

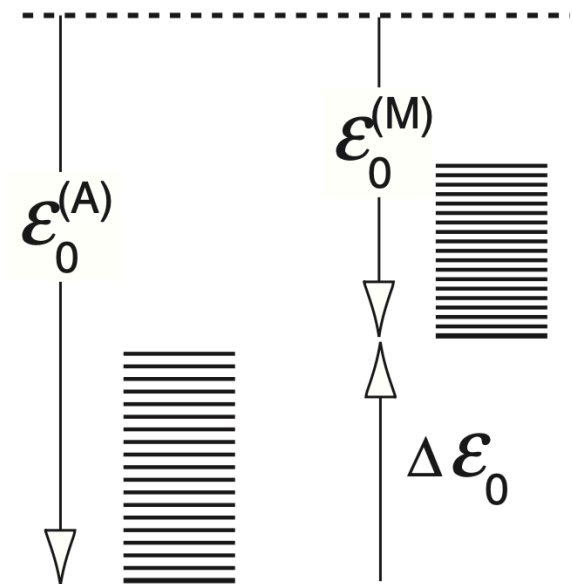
- The situation is visualized opposite. A and M both have a set of energy levels available to them, and their ground states are separated by $\Delta\epsilon_0$.



Interpretation

- *At very low temperatures, only the ground states of A and M will be occupied, so that $f_A = f_M = 1$. The equilibrium constant then just depends on $\exp(-\Delta\epsilon_0/kT)$.* In the situation shown opposite the reaction is endothermic ($\Delta\epsilon_0 > 0$) and so the equilibrium constant will be less than 1.
- *As the temperature rises the energy levels of A and M start to be occupied, leading to an increase in f_A and f_M .* This corresponds to the entropies of A and M being greater than zero, and hence the entropy change of the reaction being finite.

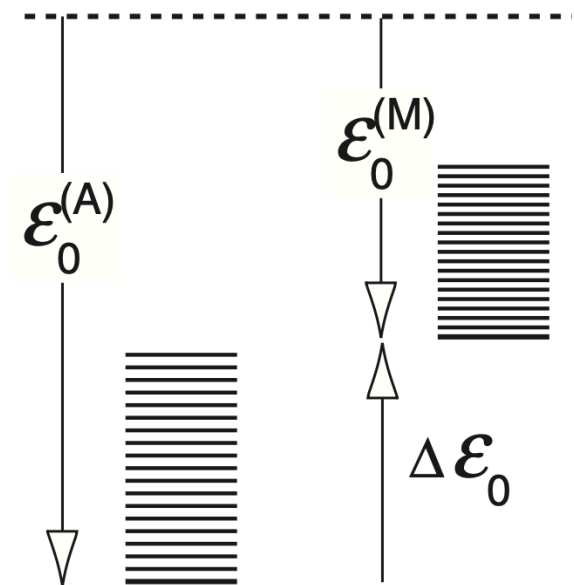
$$K_c = \frac{f_M}{f_A} \exp(-\Delta\epsilon_0/kT)$$



Interpretation

- *If the energy levels of M are more densely packed than those of A, the result will be that $f_M > f_A$.* If the difference is sufficiently large, the equilibrium constant could exceed 1 despite the term $\exp(-\Delta\epsilon_0/kT)$ being less than 1 in this case; this scenario is one in which the entropy change favours M over A. The effect of the entropy terms becomes larger the higher the temperature and, in addition, as the temperature rises the term in $\exp(-\Delta\epsilon_0/kT)$ tends towards 1, allowing the factor f_M / f_A to dominate.

$$K_c = \frac{f_M}{f_A} \exp(-\Delta\epsilon_0/kT)$$

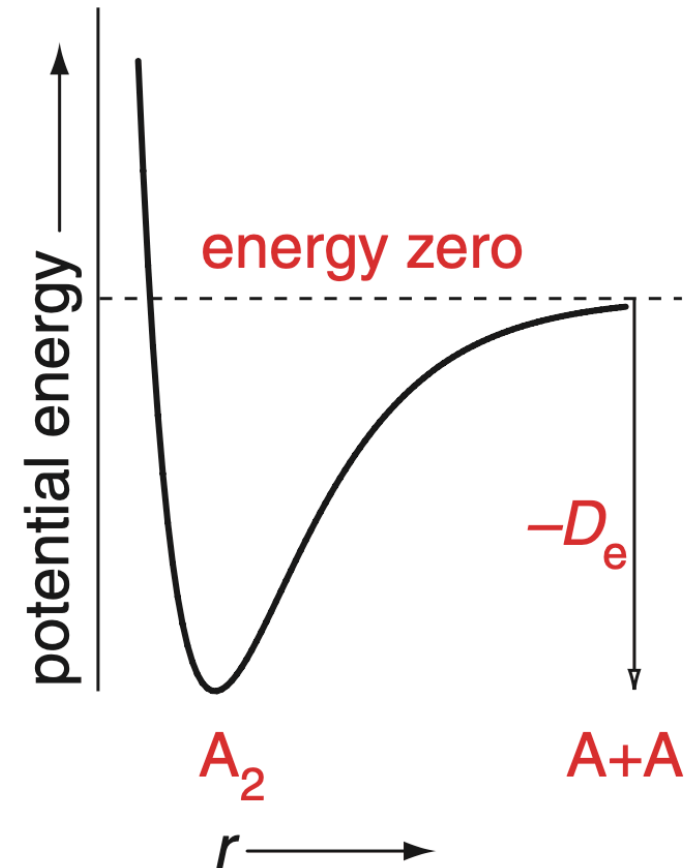


Dissociation of a diatomic

- As an example of the calculation of K_c we will consider the gas phase dissociation of a diatomic, A_2 , into two atoms:



- Shown opposite is the familiar potential energy curve for a diatomic molecule. The common energy zero is the energy of the dissociated atoms; the energy of the ground state electronic level of A_2 corresponds to the minimum in the potential energy curve and lies D_e below the energy of the atoms.



Dissociation of a diatomic

- Noting that $\Delta \nu$ is 1, the expression for K_c is therefore

$$K_c = \frac{f_A^2}{f_{A_2}} \frac{1}{c^\circ} \exp(-\Delta \epsilon_0/kT)$$

where

$$\Delta \epsilon_0 = 2\epsilon_0^{(A)} - \epsilon_0^{(A_2)} = 0 - (-D_e) = D_e$$

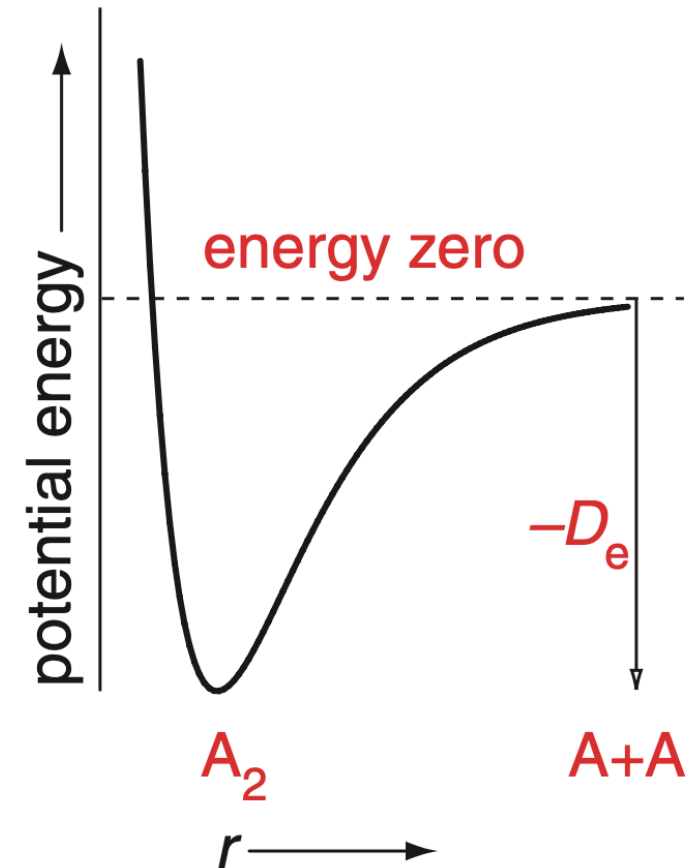
- Note that as the electronic ground state of A_2 is lower than that of the atoms, i.e. $\epsilon_0(A_2)$ is negative. The final expression is

$$K_c = \frac{f_A^2}{f_{A_2}} \frac{1}{c^\circ} \exp(-D_e/kT)$$

- It is easy to get confused with the signs here. A useful check is to look at the final expression: it says that as D_e gets larger the equilibrium constant gets smaller – which is exactly what you would expect. This sign is OK, therefore.



$$K_c = \frac{f_M^{\nu_M} f_N^{\nu_N}}{f_A^{\nu_A} f_B^{\nu_B}} \left(\frac{1}{c^\circ} \right)^{\Delta \nu} \exp(-\Delta \epsilon_0/kT)$$



Example

- Use the following data to compute K_c at 1000 K for the equilibrium:



For H_2 : $\theta_{\text{rot}} = 85.3 \text{ K}$, $\theta_{\text{vib}} = 6215 \text{ K}$ and $D_e = 36,100 \text{ cm}^{-1}$; H_2 has a closed shell so its electronic ground state is not degenerate.
For H : the ground state term symbol is $^2\text{S}_{1/2}$.

- Determine the equilibrium ratio of ortho- to para-hydrogen at 800 K. θ_{rot} for H_2 is 85.3 K.